

Course Curriculum for B Tech in Artificial Intelligence and Data Science 2023-2024 onwards

Computer Science and Engineering Department



**NATIONAL INSTITUTE OF TECHNOLOGY
DELHI**

(An autonomous Institute under the aegis of Ministry of
Education, Govt. of India.)



राष्ट्रीय प्रौद्योगिकी संस्थान दिल्ली

NATIONAL INSTITUTE OF TECHNOLOGY DELHI

(शिक्षा मंत्रालय, भारत सरकार के अधीन एक स्वायत्त संस्थान)

(An autonomous Institute under the aegis of Ministry of Education (Shiksha Mantralaya), Govt. of India)

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Course Scheme for B. Tech (AI&DS) AY 2023-24

Year	First Semester						Second Semester					
	Sub. Code	Subject Name	L	T	P	C	Sub. Code	Subject Name	L	T	P	C
I	ADLB 101	Mathematical Foundations for Data Science	3	0	0	3	ADLB 151	Probability and Statistics	3	0	0	3
	ADLB 102	Discrete Mathematics	3	1	0	4	ADBB 152	Computer Organization and Architecture	3	0	2	4
	ADBB 103	Computer Programming-I	3	0	2	4	ADBB 153	Data Structures and Algorithms	3	0	2	4
	ADBB 104	Computer Fundamentals	2	0	2	3	ADBB 154	Programming using Python	1	0	2	2
	PHLB 112	Quantum Physics	3	1	0	4	ADLB 155	System Programming	3	0	0	3
	HMLB 102	Theory and Practices of Human Ethics	1	0	0	1	CELB 101	Environmental Sciences	2	0	0	2
	HSPB 150	Holistic Health & Sports	0	0	2	1	ADPB 156	Project I	0	0	4	2
		Total	16	2	4	20	Total		13	1	12	20
	Third Semester						Fourth Semester					
II	ADBB 201	Artificial Intelligence	3	0	2	4	ADBB 251	Data Science	3	0	2	4
	ADBB 202	Database Management Systems	3	0	2	4	ADBB 252	Data Warehousing and Mining	3	0	2	4
	ADLB 203	Optimization Techniques	3	1	0	4	ADBB 253	Big Data Management	2	0	2	3
	ADBB 204	Operating Systems	3	0	2	4	ADBB 254	Machine Learning	3	0	2	4
	ADBB 205	Computer Graphics	3	0	2	4	ADLB 255	Theory of Computation	3	0	0	3
							ADPB 200	Project II	0	0	4	2
		Total	15	1	8	20	Total		14	0	12	20



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Year	Fifth Semester						Sixth Semester					
III	Sub. Code	Subject Name	L	T	P	C	Sub. Code	Subject Name	L	T	P	C
	ADBB 301	Deep Learning	3	0	2	4	ADLB 351	Social Network Analysis	3	1	0	4
	ADLB 302	Natural Language Processing	3	1	0	4	ADBB 352	Big Data Analytics	3	0	2	4
	ADBB 303	Cloud Computing	3	0	2	4	ADBB 353	Soft Computing	3	0	2	4
	ADBB 304	Image Processing and Computer Vision	3	0	2	4	ADLB XXX	Program Elective-I	3	0	0	3
	ADLB 305	Internet of Things	3	1	0	4	ADLB XXX	Program Elective -I	3	0	0	3
		Total	15	2	6	20	ADPB 300	Project - III	0	0	4	2
IV							ADPB 354	Internship (during summer break)	Credit will be given to the next Semester			
								Total	15	1	8	20
	Seventh Semester						Eighth Semester					
	ADLB 401	Cyber Security	3	0	0	3	ADPB 400	B. Tech Project (Internship inside NIT Delhi / Outside NIT Delhi)	-	-	-	16
	ADLBXXX	Program Elective-II	3	1	0	4	ADLB 451	Independent Study/ MOOC Course	3	0	0	3
	ADLBXXX	Program Elective-II	3	1	0	4	ADPB 452	Seminar	0	0	2	1
	ADBBXXX	Program Elective-III	3	0	2	4		Total	3	0	2	20
	ADBBXXX	Program Elective-III	3	0	2	4						
	ADPB 354	Internship	0	0	2	1						
		Total	12	2	6	20						

Course Code: ADBB 201	Open course (YES/NO)	HM Course (Y/N)	DC (Y/N)	DE (Y/N)	
	NO	NO	Yes	NO	
Type of course	Core				
Course Title	ARTIFICIAL INTELLIGENCE				
Course Objectives:	Gain a comprehensive understanding of Artificial Intelligence, covering its historical development, problem-solving techniques, search strategies, logical reasoning, and planning methods, with a focus on practical applications, particularly in the field of robotics. Develop essential skills to tackle complex AI challenges effectively.				
Course Outcomes	CO1: Understand the basic concepts of AI.			L1, L2	
	CO2: Apply search strategies to solve AI problems.			L3	
	CO3: Apply knowledge representation and reasoning to solve real world AI Problems.			L3	
	CO4: Explore machine learning concepts and algorithms for real world applications.			L4	
Semester	Autumn:		Spring: YES		
III	Lecture	Tutorial	Practical	Credits	Total teaching hours
Contact Hours	3	0	2	4	60
Prerequisite course code as per proposed course numbers					
Prerequisite credits					
Equivalent course codes as per proposed course and old course					
Overlap course codes as per proposed course numbers					
Text Books:					
1	Title	Artificial intelligence : A Modern Approach,			
	Author	Stuart Russell, Peter Norvig			
	Publisher	Prentice Hall			
	Edition	Fourth edition, 2020.			

Reference Book:		
2.	Title	Artificial Intelligence: A New Synthesis
	Author	Nils J. Nilsson
	Publisher	Morgan-Kaufmann, 1998.
	Edition	
3.	Title	Heuristics: Intelligent Search Strategies for Computer Problem Solving
	Author	Judea Pearl
	Publisher	Addison-Wesley Publishing Company
	Edition	1984
Content	UNIT 1 Introduction, History, Possible Approaches in AI, Automated Problem Solving Agent: Intelligent Agent & Environment, Complex Problems and AI, Shannon number, Problem Representation in AI UNIT 2 Search Strategies: Search introduction, Uninformed Search, Informed/Heuristic Search, Beyond Classical Search, Local Search, Problem Reduction, Adversarial Search, Constraint Satisfaction Problems UNIT 3 Logic and Deduction: Logical Agents, Propositional logic and Predicate Logic, First Order Logic, inference in First order Logic, Inferencing By Resolution Refutation, Classical Planning, UNIT 4 Quantifying Uncertainty, Introduction of Probability, Probabilistic Reasoning, Bayes Net, Bayesian Network, Fuzzy Logic, Decisions Theory, Utility Function, Decision Network, Markov Decision Process, Probabilistic Reasoning over time, Hidden Markov Model, Kalman filter, Markov Chain Monte Carlo UNIT 5 Learning Agent, Introduction to Machine Learning, Types of Machine Learning, Learning from experience: Reinforcement Learning, Background, Model based and Model free learning, TD and Q Learning, RL Applications, Learning from Example, Supervised learning : Introduction, Naive Bayes, Decision Tree, Perceptrons, Neural Network, Introduction to Deep Learning. AI Applications and Ethics, Ethics of AI	
Course Assessment	Continuous Evaluation 25%	
	Mid Semester 25%	
	End Semester 50%	

Lab Experiments:

Exp. No.	List of Experiments
1	Introduction to Prolog programming
2	Python Frameworks Tutorial (with Jupyter and Colab) and it's Data Structures
3	Searching in graph based problem space, exploring Uninformed search Techniques
4	Exploring Informed search Techniques (Vacuum world and Maze Problem)
5	Exploring Uninformed and Informed search Techniques (PACMAN Search Space)
6	Multi agent in a search space
7	Introduction Logical Agent and Knowledge representation using Prolog
8	Reasoning Under Uncertainty using Bayesian Learning
9	Reinforcement Learning using Q-Learning
10	Introduction to Machine Learning and Python libraries for Data Analysis (Pandas, NumPy, Matplotlib)

Course Code: ADBB 202	Open course (YES/NO)	HM Course (Y/N)	DC (Y/N)	DE (Y/N)		
	No	No	Yes	NO		
Type of course	Core					
Course Title	Database Management Systems					
Course Objectives:	1. To understand the role and functions of a database management system and its impact on the overall performance of a computer system. 2. To understand the concepts and techniques involved in ER modeling. 3. To understand the SQL commands and relational algebraic expressions for query processing. 4. To gain hands-on experience designing and implementing database management systems through programming projects and case studies.					
Course Outcomes	C01: Learn the basic concepts of Database Systems				L2	
	C02: Model the real-world systems using Entity Relationship Diagrams and convert the ER model into a relational logical schema using various mapping algorithms				L3	
	C03: Make use of SQL commands and relational algebraic expressions for query processing				L4	
	C04: Simplify databases using normalization process based on identified keys and functional dependencies and solve the atomicity, consistency, isolation, durability, transaction, and concurrency-related issues of databases				L5	
Semester		Autumn: Yes		Spring:		
V		Lecture	Tutorial	Practical	Credits	Total teaching hours
Contact Hours		3	0	2	4	60
Prerequisite course code as per proposed course numbers						
Prerequisite credits						
Equivalent course codes as per proposed course and old course						
Overlap course codes as per proposed course numbers						
Text Books:						
1		Title	Fundamentals of Database Systems			
		Author	R. Elmasri and S.B. Navathe			
		Publisher	Pearson			
		Edition	2016			
Reference Book:						

1.	Title	Database Systems Concepts
	Author	H.f.Korth and Silberschatz
	Publisher	McGraw Hill
	Edition	
2.	Title	Data Base Design
	Author	C.J. Date
	Publisher	Addison Wesley
	Edition	
3.	Title	DBM and Design
	Author	Hansen and Hansen
	Publisher	PHI
	Edition	
4.	Title	Database System
	Author	Hector Garcia-Molina, Jeff Ullman, and Jennifer Widom
	Publisher	Pearson
	Edition	2 nd Edition

Content	Unit 1 (6 hours) Introduction - General introduction to database systems; Database - DBMS distinction, approaches to building a database, data models, database management system, three-schema architecture of a database, challenges in building a DBMS, various components of a DBMS. Unit 2 (8 hours) Database design and ER Model:- Overview, ER-Model, Constraints, ER-Diagrams, ERD Issues, weak entity sets, Codd's rules, Relational Schemas, Introduction to UML Relational database model: Logical view of data, keys, integrity rules. Relational Database design: features of good relational database design, atomic domain, and Normalization (1NF, 2NF, 3NF, BCNF). Unit 3 (6 hours) Relational algebra: introduction, Selection and projection, set operations, renaming, Joins, Division, syntax, semantics. Operators, grouping and ungrouping, relational comparison. Calculus: Tuple relational calculus, Domain relational Calculus, calculus vs algebra, computational capabilities. Unit 4 (8 hours) SQL - Introduction, data definition in SQL, table, key and foreign key definitions, update behaviors. Querying in SQL - basic select-from-where block and its semantics, nested queries- correlated and uncorrelated, notion of aggregation, aggregation functions group by and having clauses, embedded SQL. Data Storage and Indexes - file organizations, primary, secondary index structures, various index structures - hash-based, dynamic hashing techniques, multi-level indexes, and B+ trees. Unit 5 (8 hours) Transaction management and Concurrency control: Transaction processing and Error recovery - concepts of transaction processing, ACID properties, and serializability concurrency control, Lock based concurrency control (2PL, Deadlocks), Time stamping methods, optimistic methods, and database recovery management. Error recovery and logging, undo, redo, undo-redo logging, and recovery methods.
Course Assessment	Continuous Evaluation 25%
	Mid Semester 25%
	End Semester 50%

Lab Experiments:

Exp. No.	List of Experiments
1	Library Management system (File Handling)
2	Introduction to SQL ● Installation of SQL-Server ● SQL data definition ● Constraints in SQL ● Schema change Statement
3	Basic SQL Queries
4	Complex SQL Queries-1 ● Nested Queries

	• Correlated Nested Queries • EXISTS Function in SQL • Aggregation Function
5	Complex SQL Queries-2 • Joined Tables • Aggregate Functions
6	Complex SQL Queries-3 • Grouping • EXISTS and UNIQUE functions • Aggregate Functions
7	Entity-Relationship Diagram from Case Study
8	Normalization of the Case Study
9	Webpage Connectivity with SQL Server Using XAMPP- 1
10	Webpage Connectivity with SQL Server Using XAMPP- 2
11	Mini DBMS Project
12	Mini DBMS Project

Course no: ADLB 203	Open course (YES/ NO)	HM Course (Y/N)	DC (Y/N)		DE (Y/N)
	NO	NO	NO		NO
Type of course	Elective				
Course Title	Optimization Techniques				
Course Coordinator					
Course objectives:	This course aims to cover the concepts of optimization methods and algorithms developed for solving various types of optimization Problems. To apply the mathematical results and numerical techniques of Optimization theory to various Engineering and Analytics problems. Explain the theoretical workings of the graphical, simplex and analytical methods for making effective decision on variables so as to optimize the objective function.				
Course Outcomes:	C01: To understand the fundament of Linear Programming and Dynamic Programming.		L1,L3		
	C02: Enumerate fundamentals of Integer programming technique and apply different techniques to solve various optimization problems arising from engineering areas.		L1,L2		
	C03: Identify appropriate optimization method to solve complex problems involved in various industries.		L1,L2,L4		
	C04: To understand the graphical, simplex and analytical methods for making effective decision.		L2,L5		
Semester	Autumn: Yes		Spring: Yes		
	Lecture	Tuto rial	Practic al	Cred its	Total teaching hours
Contact Hours	3	1	0	4	48
Prerequisite course code as per proposed course numbers	Linear algebra, Calculus				
Prerequisite credits	NIL				
Equivalent course codes as per proposed course and old course	NIL				

Overlap course codes as per proposed course Numbers		NIL				
Text Books:						
1.		Title	An Introduction to Optimization			
		Author	Edwin K.P. Chong, Stanislaw H. Zak,			
		Publisher	Wiley			
		Edition	4 th			
Reference Book:						
1.		Title	Convex Optimization			
		Author	Stephen Boyd			
		Publisher	LievenVandenberghe			
		Edition	3 rd			
2.		Title	Modern Optimization with R (Use R)			
		Author	Paulo Cortez			
		Publisher	Springer			
		Edition	2014			
Content		Unit 1 Preliminaries: Proofs, Vector Spaces and Matrices, Linear Transformations, Eigenvalues and Eigenvectors, Orthogonal Projections, Quadratic Forms, Matrix Norms, Concepts from Geometry, Elements of Calculus Unit 2 Unconstrained Optimization: Basics of Set Constrained and Unconstrained Optimization, One Dimensional Search Methods, Golden Section Search, Fibonacci Search, Newton's Method, Secant Method, Solving $Ax = b$ Unit 3 Linear Programming: Introduction to Linear Programming, Simplex Method, Duality Unit 4 Nonlinear Constrained Optimization: Problems with Equality Constraints, Problems with Inequality Constraints, Karush Kuhn Tucker Condition, Convex Optimization Problems, Unit 5 Algorithms for Constrained Optimization: Projections, Project gradient methods, Penalty methods.				
Course Assessment		Continuous Evaluation 25% Mid Semester 25% End Semester 50%				

Course no: ADBB204	Open course (YES/NO)	HM Course (Y/N)	DC (Y/N)	DE (Y/N)	
Type of course	Program Core				
Course Title	Operating Systems				
Course objectives:	• To understand the role and functions of an operating system and its impact on the overall performance of a computer system. • To understand the concepts and techniques involved in process management, such as process creation, scheduling, interprocess communication, and synchronization. • To understand the concepts and techniques involved in memory management, such as virtual memory, swapping, paging, and segmentation. • To gain hands-on experience with the design and implementation of operating systems through programming projects and case studies.				
POs					
Semester	Autumn: Yes		Spring:		
	Lecture	Tutorial	Practical	Credits	Total teaching hours
Contact Hours	3	0	2	4	36
Prerequisite course code as per proposed course numbers	NIL				
Prerequisite credits	NIL				
Equivalent course codes as per proposed course and old course	NIL				
Overlap course codes as per proposed course numbers	NIL				
Text Books:					
1	Title	Operating System concepts			
	Author	Abraham Silberschatz, Peter B. Galvin, Greg Gagne			
	Publisher	Addison-Wesley			
	Edition	Eighth edition, 2017			
Reference Book:					
1	Title	Modern Operating Systems			
	Author	Andrew Tanenbaum			
	Publisher	Prentice Hall			
	Edition				
2	Title	Operating Systems			
	Author	William Stallings			
	Publisher	Prentice Hall			
	Edition				

3	Title	An introduction to operating systems
	Author	Harvey M. Deitel
	Publisher	Addison-Wesley
	Edition	
Content	Unit – 1 (5 Hours) Basics: Operating System Functionalities, Types of Operating Systems, Computer Architecture support to Operating Systems. Unit – 2 (8 Hours) Process Management: Threads, Process Scheduling - Uniprocessor scheduling algorithms, Multiprocessor and Real-time scheduling algorithms, Process Synchronization - Peterson's Solution, Bakery. Algorithm, Hardware Support to Process Synchronization, Semaphores, Critical Regions, Monitors - Deadlock prevention, deadlock avoidance, and Detection and Recovery - Bankers Algorithm. Unit – 3 (7 Hours) Memory Management: Segmentation and space allocation, Basics of linking and loading, Demand Paging, Page replacement algorithms, Analysis of page allocation policies, Thrashing- Working Set. Unit – 4 (8 Hours) File Systems: Contiguous, Sequential, and Indexed Allocation, File system interface, File System implementation; Case study of Unix File system, Mounting and Unmounting files systems; Network File systems. Unit – 5 (8 Hours) I/O System: Disk Scheduling, Device drivers - block and character devices, streams, Character and Block device switch tables. Protection and Security - Accessibility and Capability Lists.	
Course Outcomes:	• CO1: Understanding of the fundamental concepts, design principles, and implementation techniques of modern operating systems • CO2: Ability to design, implement, and evaluate process management, memory management, file system management, and input/output management algorithms • CO3: Ability to understand and implement distributed systems, such as client-server systems, distributed file systems, and distributed operating systems. • CO4: Hands-on experience designing and implementing operating systems through programming projects and case studies.	
Course Assessment	Continuous Evaluation 25%	
	Mid Semester 25%	
	End Semester 50%	

Lab Experiments:

Exp. No.	Experiments
1.	Basics of Unix Commands
2.	Implementation of Process Related System Calls (Fork).
3.	Implementation of System Calls (Open, Read, Write, and Close) for File Management
4.	Implementation of Process Synchronization
5.	Implementation of Memory Management Using Address Translation
6.	Implementation of FIFO Page Replacement Algorithms
7.	Implementation of LRU Page Replacement Algorithms
8.	Implementation of First Come First Serve and Shortest Job First Scheduling Algorithm
9.	Implementation of Priority and Round Robin CPU Scheduling Algorithm
10.	Implementation of Banker's Algorithm.
11.	Implementation of Sleeping Barber Problem in Process Synchronization
12.	Implementation of Algorithm for Deadlock Detection

Course no: ADBB205	Open course (YES/NO)		HM Course (Y/N)	DC (Y/N)	DE (Y/N)
Type of course	Program Core				
Course Title	Computer Graphics				
Course objectives:	- Emphasize basic principles needed to design, use, and understand computer graphics systems. - Discuss both hardware and software components of graphics systems. - Implementation of Graphics Algorithms				
POs					
Semester		Autumn: Yes		Spring: No	
	Lecture	Tutorial	Practical	Credits	Total teaching hours
Contact Hours	3	0	2	4	36
Prerequisite course code as per proposed course numbers	NIL				
Prerequisite credits	NIL				
Equivalent course codes as per proposed course and old course	NIL				
Overlap course codes as per proposed course numbers	NIL				
Text Books:					
1	Title	Computer graphics C Version			
	Author	Donald Hearn and M. Pauline Baker			
	Publisher	Pearson			
	Edition	2nd edition, 2014			
Reference Book:					
1	Title	Computer Graphics-Principles and Practice			
	Author	J. D. Foley, A. Van Dam, S. K. Feiner and J. F. Hughes			
	Publisher	Pearson			
	Edition	2nd edition, 2003			
2	Title	Procedural Elements of Computer graphics			
	Author	David F. Rogers			
	Publisher	McGraw Hill			
	Edition	2nd edition, 1998			

3	Title	Computer Graphics: A programming approach
	Author	Steven Harringtons
	Publisher	McGraw Hill
	Edition	2nd edition, 1987
Content	Unit – 1 (5 Hours) OVERVIEW OF GRAPHICS SYSTEM: Input devices, Output primitives, Video display devices, Raster and vector graphics, linecircle-ellipse generating algorithm, filled area primitives Unit – 2 (8 Hours) TRANSFORMATIONS AND PROJECTIONS: 2D Transformation, 3D transformation, Parallel projection, Perspective projection. Unit – 3 (7 Hours) LINE CLIPPING: Cohen-Sutherland, Liang Barsky, Polygon clipping: Sutherland Hodgeman &Weiler-Atherton polygon clipping. Unit – 4 (8 Hours) CURVES & SURFACES: Conics-Parametric forms for circle, ellipse, parabola; Bezier Curves, Bernstein polynomials, Convex Hull property, B-Spline Curves: Knot vectors uniform and open uniform curves, Periodic B-splines, Uniform B-splines, Non-uniform, rational B-splines, subdividing curves. Unit – 5 (8 Hours) HIDDEN SURFACE REMOVAL: Hidden Surface Removal: Back face removal, Floating Horizon method for curved objects, Z-Buffer or depth buffer algorithm, Painter’s algorithm(Depth sorting method), Binary space partitioning trees, Scan line algorithm, Warnock’s algorithm (Area subdivision method).	
Course Outcomes:	CO1: Demonstrate understanding of the basics of Computer Graphics (L2). CO2: Develop understanding and underlying techniques and algorithms of Graphics Primitives, Display Methods and Visible surface detection concepts (L3). CO3: Develop understanding of frequency domain processing techniques. (L3) CO4: Develop understanding of modelling techniques used to restore images (L3) CO5: Develop understanding of color image processing and compressing techniques (L3)	
Course Assessment	Continuous Evaluation 25%	
	Mid Semester 25%	
	End Semester 50%	

Lab Experiments:

Exp. No.	Experiments
1.	Digital Differential Analyzer Algorithm
2.	Bresenham's Line Drawing Algorithm
3.	Midpoint Circle Generation Algorithm
4.	Ellipse Generation Algorithm
5.	Creating various types of texts and fonts
6.	Creating two dimensional objects
7.	Two Dimensional Transformations
8.	Coloring the Pictures
9.	Three Dimensional Transformations
10.	Curve Generation
11.	Simple Animations using transformations
12.	Key Frame Animation

Course no: ADBB 251	BSC (YES/ NO)	ESC (YES/ NO)	HSC (YES/ NO)	PC (YES/ NO)	PE (YES/ NO)	IN-IS-SP-MP (YES/ NO)
	NO	NO	NO	YES	NO	NO
Type of course	Program Core					
Course Title	DATA SCIENCE					
Course objectives:	The purpose of this course is to understand the foundations of Data Science and its applications. This course will equip students with the skills to preprocess, analyze, and interpret data, and further will introduce machine learning techniques for predictive analytics. The main aim would be to provide hands-on experience in working with real-world datasets.					
POs						
Semester		Autumn:		Spring: Yes		
IV		Lecture	Tutorial	Practical	Credits	Total teaching hours
Contact Hours		3	0	1	4	36
Prerequisite course code as per proposed course numbers		NIL				
Prerequisite credits		NIL				
Equivalent course codes as per proposed course and old course		NIL				
Overlap course codes as per proposed course numbers		NIL				
Text Books:						
1		Title	Introducing Data Science			
		Author	Davy Cielen, Arno D. B. Meysman, Mohamed Ali			
		Publisher	Ebury Press			
		Edition	First edition, 2023			
Reference Book:						
1		Title	Introduction to Data Science: Practical Approach with R and Python			
		Author	B. Uma Maheswari, R. Sujatha			
		Publisher	Wiley			
		Edition	First edition, 2021			
2		Title	The Elements of Statistical Learning			
		Author	Trevor Hastie, Robert Tibshirani, Jerome Friedman			
		Publisher	Springer			
		Edition	Second edition, 2009			

3	Title	Python for Data Analysis
	Author	Wes McKinney
	Publisher	Shroff/O'Reilly
	Edition	Third edition, 2022
Content	Unit -1: Foundational Mathematics for Data Science (6 Hours) Linear algebra; vectors; matrices; product of matrix & vector; rank; null space; solution of over-determined set of equations and pseudo-inverse; distance; projections; eigenvalue decomposition; statistical modeling/descriptive statistics, notion of probability (random variables), distributions, mean, variance, covariance, covariance matrix, introduction to hypothesis testing, One-Tailed and Two-Tailed Test, Type 1 and Type 2 errors, Test Statistic. Unit-2: Python Programming for Data Science (8 Hours) Introduction to python; variables and datatypes; operators; sequence data; control structures and functions; reading data; most widely used library for python: Pandas, NumPy, SciPy, Matplotlib Unit –3: Data Cleaning and Exploration (8 Hours) Data analysis: types of data (Structured, unstructured, and semi-structured); Data preprocessing: Missing values, outliers, normalization, and standardization; Feature engineering: Encoding, scaling, and transformation, Singular value decomposition (SVD,)Principal component analysis(PCA) Unit –4: Data modeling (7 Hours) Predictive modeling; Linear regression, multiple linear regression; cross-validation; classification using logistic regression; classification using kNN and k-means clustering. Unit –5: Model optimization (7 Hours) Typology of data science problems and a solution framework; optimization techniques like multivariate optimization – unconstraint, equality constraint, unequality constraint; Gradient descent and cost functions.	

Course Outcomes	• Understand and apply data preprocessing techniques. • Implement machine learning algorithms. • Utilize statistical methods for data interpretation. • Design visualizations to effectively communicate findings.
Course Assessment	Continuous Evaluation 25%
	Mid Semester 25%
	End Semester 50%

Lab Experiments:

Exp. No.	List of Experiments
1	Learn to install and use of various libraries used for data handling in python a. Install Anaconda and Jupyter Notebook b. Install Numpy, Pandas, SciPy, Matplotlib packages and learn basis operations on datasets using these packages c. Load data from file stored in local computer and visualize it

2	Implement eigenvalue decomposition on provided data sets: a. Using Numpy library b. Using SciPy library
3	Implement Singular Value Decomposition (SVD) on given datasets.
4	Analyze the relationship between the given data series by calculating their Covariance and Correlation. Create scatter plots to visualize the data and include a title displaying the Covariance and Correlation values.
5	Verify if given dataset is normally distributed or not by following methods: a. Visual method: create Histogram, Create a Q-Q plot b. Shapiro-Wilk Test c. Kalmogorov-Smirnov Test
6	State hypothesis and Perform following T statistics tests on given datasets a. One sample T-test b. Independent sample T-test c. Paired sample T-test
7	State hypothesis and Perform following F statistics tests on given datasets a. One way ANOVA test b. Two way ANOVA test
8	Perform following data preprocessing techniques on the given dataset. a. Handling missing data, non-relevant, noisy and outlier data b. Handling Categorical variables in data c. Scaling and normalization d. Handling Unbalanced data
9	Perform Principal Component Analysis (PCA) on given dataset.
10	Model the given datasets using linear regression, logistic regression, log-linear regression, and multiple regressions.
11	Implement the Decision Tree Algorithm on the provided dataset.
12	Apply the KNN algorithm to a given dataset.

Course no: ADBB 252	BSC (YES/ NO)	ESC (YES/ NO)	HSC (YES/ NO)	PC (YES/ NO)	PE (YES/ NO)	IN-IS-SP-MP (YES/ NO)
	NO	NO	NO	YES	NO	NO
Type of course	Program Core					
Course Title	Data Mining and Warehousing					
Course objectives:	The purpose of this course is to know the fundamental concepts of Data Mining and Warehousing, explore tools and practices for working with Data and apply analytics on structured and sunstructured data.					
POs						
Semester		Autumn:		Spring: Yes		
IV		Lecture	Tutorial	Practical	Credits	Total teaching hours
Contact Hours		3	0	2	4	36 + 22
Prerequisite course code as per proposed course numbers		NIL				
Prerequisite credits		NIL				
Equivalent course codes as per proposed course and old course		NIL				
Overlap course codes as per proposed course numbers		NIL				
Text Books:						
1		Title	Data Mining Concepts and Techniques			
		Author	Jiawei Han and Micheline Kamber			
		Publisher	Morgan Kaufmann			
		Edition	2011			
2		Title	Data Mining: Practical Machine Learning Tools and Techniques			
		Author	Eibe Frank and Ian H. Witten			
		Publisher	Morgan Kaufmann			
		Edition	Third Edition, 2011			
3		Title	Introduction to Data Mining			
		Author	Pang-Ning Tan Michael Steinbach Vipin Kumar			
		Publisher	Pearson			
		Edition	Global edition, 2019			
Reference Books:						
1		Title	Database Concepts			

	Author	Abraham Sibertschatz, Henry F. Korth and S. Sudarshan
	Publisher	McGraw Hill
	Edition	Seventh Edition, 2019
Content	Unit -1: Introduction to Data Mining and Data Warehouse (8 Hours) Design Guidelines for Data Warehouse Implementation, Multidimensional Models, OLAP - Introduction, Characteristics, Architecture, Multidimensional view, Efficient Processing of OLAP Queries, OLAP Server Architecture, ROLAP versus MOLAP Versus HOLAP and Data Cube, Data Cube Operations, Data Cube Computation. Motivation for data mining, Introduction to data mining system, Data mining functionalities, KDD, Data object and attribute types, Statistical description of data, Issues and Applications. Unit-2: Machine Learning Concepts and Approaches (6 Hours) Supervised Learning Framework, Concepts & Hypothesis, Training & Learning, Boolean Functions and Formulae, Monomials, Disjunctive Normal Form & Conjunctive Normal Form, A Learning Algorithm for Monomials. Unit –3: Data Preparation and Mining Association Rules (8 Hours) Data cleaning, Data integration and transformation, Data reduction, Data discretization and Concept Hierhy Generation, Data mining primitives Frequent patterns, Market basket analysis, Frequent itemsets, closed itemsets, association rules, Types of association rule (Single dimensional, multidimensional, multilevel, quantitative), Finding frequent itemset (Apriori algorithm, FP growth), Generating association rules from frequent itemset, Limitation and improving Apriori, From Association Mining to Correlation Analysis, Lift. Unit –4: Classification and Prediction and Cluster Analysis (8 Hours) Issues regarding Classification & Prediction, Classification by Decision Tree Induction, Bayesian Classification, Classification by Back Propagation, k-Nearest Neighbour Classifiers, Genetic Algorithms, Rough Set & Fuzzy Set Approaches. Unit –5: Cluster Analysis (6 Hours) Types of Data in Clustering Analysis, Categorization of Major Clustering Methods, Hierarchical Methods, Density-based methods, Grid-based methods, Grid-based methods, Model-based Clustering Method.	
Course Outcomes	● Explain the concept and significance of Data Mining (L2). ● Explore Recent Trends in Data Mining such as Web Mining, Spatial-Temporal Mining (L2). ● Analyze different mining algorithms and clustering techniques for Data Analytics (L3). ● Design and Develop a Data Warehouse for an organization (L6).	
Course Assessment	Continuous Evaluation 25%	
	Mid Semester 25%	
	End Semester 50%	

Lab Experiments:

Exp. No.	List of Experiments
1	Load Data from heterogenous sources including text files into a predefined warehouse schema.
2	Design a data mart for a bank to store the credit history of customers in a bank, Use this credit profiling to process future loan applications.
3	Feature Selection and Variable Filtering (For very large data sets).
4	Association Mining in large data sets
5	Interactive Drill-Down, Roll up, Slice and Dice Operations.
6	Generalized EM and k-Means Cluster Analysis
7	Generalized Additive Models (GAM).
8	General Classification Regression Tress (GTrees)
9	General CHAID (Chi-square Automatic Interaction Detection) Models
10	Interactive Classification and Regression Trees.
11	Goodness of Fit Computations

Course Code: ADBB 253	BSC (YES/NO)	ESC (YES/NO)	HSC (YES/NO)	PC (YES/NO)	PE (YES/NO)	IN-IS-SP-MP (YES/NO)
	NO	NO	NO	YES	NO	NO
Type of course	Program Core					
Course Title	BIGDATA MANAGEMENT					
Course objectives:	This course introduces the fundamentals of Big Data, its applications, and challenges and provides a comprehensive understanding of data storage, processing, and management techniques. Main aim for the course is to understand the tools and frameworks for Big Data analytics, data streaming, batch processing, and visualization technique and to integrate Big Data management concepts with real-world applications and case studies.					
POs						
Semester	Autumn:		Spring: Yes			
IV	Lecture	Tutorial	Practical	Credits	Total teaching hours	
Contact Hours	2	0	1	3	30	
Prerequisite course code as per proposed course numbers	NIL					
Prerequisite credits	NIL					
Equivalent course codes as per proposed course and old course	NIL					
Overlap course codes as per proposed course numbers	NIL					
Text Books:						
1	Title	Big Data and Analytics				
	Author	Seema Acharya, Subhashini Chellappan				
	Publisher	Wiley				
	Edition	Second edition, 2019				
Reference Book:						
1	Title	Hadoop: The Definitive Guide				
	Author	Tom White				
	Publisher	O'Reilly				
	Edition	First edition, 2012				
2	Title	Big Data: Principle and best practices of scalable real-time data systems				
	Author	James Warren, Nathan Marz				
	Publisher	Manning Publications				
	Edition	First edition, 2015				
3	Title	MongoDB: The Definitive Guide				
	Author	S. Bradshaw, E. Brazil, K. Chodorow				

		Publisher	Shroff/O'Reilly
		Edition	Third edition, 2020
Content	Unit –1 (6 Hours) Introducing to Big Data – Characteristics and evolution of Big Data (3Vs: Volume, Velocity, Variety); Applications of Big Data: Challenges in managing Big Data; Big Data vs Traditional data. Unit-2 (8 Hours) Big Data storage; Distributed File Systems: Introduction to HDFS (Hadoop Distributed File System); Key-value storage systems: NoSQL Databases (MongoDB); Comparison: SQL vs NoSQL; Data Models: Document, Columnar, Graph-Based Systems. Unit –3 (4 Hours) Big Data processing; Batch Processing: Introduction to Hadoop and MapReduce; working mechanism of MapReduce, YARN; real-time processing: introduction to Apache Spark: RDDs (Resilient Distributed Datasets), SparkSQL and DataFrames; Data Stream Processing: Kafka, Apache Flink (Overview). Unit –4 (4 Hours) Big Data Analytics; basics of Big Data Analytics: data preprocessing for Big Data, visualization tools (Tableau, PowerBI, Python Libraries). Unit –5 (4 Hours) Big Data security and privacy: Data Security in Distributed Systems; Privacy Challenges in Big Data; Data Governance and Compliance (GDPR, HIPAA, etc.); case studies and applications in Big Data applications (IoT, E-Commerce, etc.).		
Course Outcomes	● Analyze and differentiate between traditional and Big Data technologies. ● Implement data processing pipelines using Big Data tools. ● Apply advanced storage and retrieval techniques for Big Data. ● Work with distributed systems and Big Data frameworks like Hadoop and Spark. ● Design and optimize Big Data solutions for real-world applications.		
Course Assessment	Continuous Evaluation 25%		
	Mid Semester 25%		
	End Semester 50%		

Lab Experiments:

Exp. No.	List of Experiments
1	To install and configure Hadoop (HDFS, MapReduce, YARN) on a local machine, VM or cluster, and understand the basic setup of the Hadoop ecosystem.
2	To perform basic HDFS operations such as uploading, retrieving, and managing large files stored in a distributed file system after the Hadoop installation.
3	To install Apache Hive on top of the Hadoop ecosystem and configure it for SQL-like query execution on big data.
4	To write and execute a basic MapReduce job that processes a large dataset on the Hadoop cluster, showcasing parallel processing.
5	To install and configure Apache Spark for distributed data processing, and integrate it with Hadoop.
6	To perform data processing tasks using Apache Spark, such as RDD operations, transformations, and aggregations on large datasets.
7	To install and configure Apache HBase, a NoSQL database, for handling large-scale real-time data storage and retrieval.
8	To perform data storage and retrieval operations in HBase, and analyze the use of HBase for real-time, large-scale data processing.
9	To install and configure Apache Impala for fast SQL querying on large datasets stored in Hadoop.
10	To execute complex SQL queries on large datasets using Apache Impala, and apply query optimization techniques for improved performance.

Course Code: ADBB 254	Open course (YES/NO)	HM Course (Y/N)	DC (Y/N)	DE (Y/N)		
	NO	NO	Yes	NO		
Type of course	Core					
Course Title	MACHINE LEARNING					
Course Objectives:	Gain a comprehensive understanding of Artificial Intelligence, covering its historical development, problem-solving techniques, search strategies, logical reasoning, and planning methods, with a focus on practical applications, particularly in the field of robotics. Develop essential skills to tackle complex AI challenges effectively.					
Course Outcomes	CO1: Understanding popular ML algorithms with their associated mathematical foundations.			L1, L2		
	CO2: Explore role of data in the future of computing, and also in solving real-world problems using machine learning algorithms.			L3		
	CO3: Apply knowledge representation and reasoning to solve real world Machine Learning Problems.			L3		
	CO4: Explore machine learning concepts and algorithms for real world applications.			L4		
Semester		Autumn:		Spring: YES		
IV		Lecture	Tutorial	Practical	Credits	Total teaching hours
Contact Hours		3	0	2	4	36
Prerequisite course code as per proposed course numbers						
Prerequisite credits						
Equivalent course codes as per proposed course and old course						
Overlap course codes as per proposed course numbers						
Text Books:						
1		Title	Machine Learning			
		Author	Tom M. Mitchell			
		Publisher	Prentice Hall			
		Edition	Fourth edition, 2020.			
Reference Book:						
2.		Title	Machine Learning			
		Author	Saikat Dutt, S. Chjandramouli, Das			
		Publisher	Pearson			
		Edition				
3.		Title	Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow			
		Author	Aurélien Géron			
		Publisher	O'Reilly Media			

		Edition	2nd
Content	UNIT 1: Introduction to Machine Learning (8 Hours) What is machine learning? Types of machine learning– supervised, unsupervised, semi-supervised and reinforcement learning, machine learning activities, applications of machine learning. UNIT 2: Model Preparation, Evaluation and feature engineering (8 Hours) Types of data in machine learning, Exploring structure of data, Data pre-processing, Model selection and training, Evaluating machine learning algorithms and performance enhancement of models. What is feature engineering?, Feature transformation, Feature subset selection. Principal component analysis. UNIT 3: Supervised Learning (8 Hours) Introduction of regression, Regression algorithms: Simple linear regression, Multiple linear regression, Polynomial regression model, Logistic regression, Maximum likelihood estimation. Introduction of supervised learning, Classification model and learning steps, Classification algorithms: Naïve Bayes classifier, k-Nearest Neighbour (kNN), Decision tree, Support vector machines, Random Forest. UNIT 4: Unsupervised Learning (8 Hours) Unsupervised Learning: Introduction of unsupervised learning, Unsupervised vs supervised learning, Application of unsupervised learning, Clustering and its types, Partitioning method: k-Means and K-Medoids, Hierarchical clustering, Density-based methods – DBSCAN. UNIT 5: Advance topics in Machine Learning (6 Hours) Introduction to deep learning, overview of reinforcement learning, Case-study of ML applications: Image recognition, speech recognition, Email spam filtering, Online fraud detection and other.		
Course Assessment	Continuous Evaluation 25%		
	Mid Semester 25%		
	End Semester 50%		

Lab Experiments:

Exp. No.	List of Experiments
1	Write a python program to import and export data using Pandas library functions.
2	Demonstrate various data pre-processing techniques for a given dataset.
3	Write a Python program to demonstrate various Data Visualization Techniques.
4	Implement Simple and Multiple Linear Regression Models.
5	Develop Logistic Regression Model for a given dataset.
6	Develop Decision Tree Classification model for a given dataset and use it to classify a new sample.
7	Implement Naïve Bayes Classification in Python.
8	Build KNN Classification model for a given dataset.
9	Write a python program to implement K-Means clustering Algorithm.
10	Implement Random forest ensemble method on a given dataset.

Couse No: ADLB 255	Open Course (Yes/No)	HM Course (Y/M)	DC (Y/N)	DE (Y/N)	
Type of Course	Core				
Course Title	THEORY OF COMPUTATION				
Course Objectives:	1. To give an overview of the theoretical foundations of computer science from the perspective of formal languages				
	2. To illustrate finite state machines to solve problems in computing				
	3. To explain the hierarchy of problems arising in the computer sciences.				
	4. To familiarize Regular grammars, context frees grammar.				
POs	1. To use basic concepts of formal languages of finite automata techniques				
	2. To Design Finite Automata’s for different Regular Expressions and Languages				
	3. To Construct context free grammar for various languages				
	4. To solve various problems of applying normal form techniques, push down automata and Turing Machines				
Semester	Autumn:		Spring:Yes		
IV	Lecture	Tutorial	Practical	Credits	Total Teaching Hours
Contact Hours	3	0	0	3	36
Text Books:					
1.	Title	Introduction to Automata Theory, Languages and Computation			
	Author	J. E. Hopcroft R. Motwani and J. D. Ullman			
	Publisher	Addison Wesley			
	Edition	3rd Edition, 2006			
Reference Books					
1.	Title	Introduction to the Theory of Computation			
	Author	M. Sipser			
	Publisher	Thomson			
	Edition	2001			
2.	Title	Elements of Theory of Computation			
	Author	C. H. Papadimitriou, H. Lewis			
	Publisher	Prentice Hall			
	Edition	1981			

Content	Unit-1: Regular Languages (10 Hours) Introduction to Computing, Mathematical model of computing, decision problems, set membership problems, Notion of a formal language, DFAs and notion for their acceptance, informal and then formal definitions, Class of regular languages, Closure of the class under complementation, union and intersection, Strategy for designing DFAs, Pumping lemma for regular languages, NFAs, conversion of equivalent DFAs of NFAs. NFAs with epsilon transitions, Regular expressions, Closure properties of and decision problems for regular languages, Mealy Machine and Moore Machines. Unit – 2: Context free languages (6 Hours) Notion of grammars and languages generated by grammars. Equivalence of regular grammars and finite automata. Context free grammars and their parse trees. Pushdown automata (PDAs): deterministic and nondeterministic. Language acceptance by final states and by empty stack. PDAs and CFGs, epsilon productions, unit productions from CFGs. Chomsky hierarchy. Pumping lemma for CFLs and its use. Closure properties of CFLs. Decision problems for CFLs. Unit – 3: Turing machines, Regular Expression languages (8 Hours) Turing machines (TMs), their instantaneous descriptions. Language acceptance by TMs. Hennie convention for TM transition diagrams. Church-Turing hypothesis and its foundational implications. recursive languages. non-recursive enumerable languages. Unit – 4: Undecidability (6 Hours) Notion of undecidable problems. Universal language and universal TM. Separation of recursive and r.e. classes. Notion of reduction. Some undecidable problems of TMs. Rice's theorem. Undecidability of Post's correspondence problem (PCP), some simple applications of undecidability of PCP Unit – 5: Intractability (6 Hours) Notion of tractability/feasibility. The classes NP and co-NP, their importance. Polynomial time many-one reduction. Completeness under this reduction. Cook-Levin theorem: NP-completeness of propositional satisfiability, other variants of satisfiability. NP-complete problems from other domains.
Course Assessment	Continuous Evaluation 25%
	Mid Semester 25%
	End Semester 50%

Syllabus for B. Tech AI &DS (3rd Year – V and VI Semesters)

Course Code: ADBB 301	PC (YES/ NO)	PE (YES/ NO)	OE (YES/ NO)	AS (YES/ NO)	HM (YE S/ NO)	ST-IS-PR (YES/ NO)	AE (YES / NO)
	YES	NO	NO	NO	NO	NO	NO
Type of course	Program Core						
Course Title	DEEP LEARNING						
Course Objectives	To introduce the fundamentals and advanced concepts of deep learning, enabling students to design and apply neural network models including CNNs, RNNs, transformers, and generative models for solving real-world problems in vision, language, and time series domains using frameworks like TensorFlow and PyTorch.						
Course Outcomes	CO1: Understand the fundamental concepts of deep learning, including artificial neural networks, activation functions, loss functions, and optimization techniques.					L1, L2, L3	
	CO2: Apply convolutional and recurrent neural network architectures to solve real-world tasks such as image classification, object detection, and natural language processing.					L4, L5, L6	
	CO3: Analyze the working of advanced models like Transformers, Vision Transformers (ViT), and Swin Transformers for both vision and language-related applications.					L4, L5, L6	
	CO4: Design generative and time series models using autoencoders, GANs, diffusion models, and deep learning-based forecasting techniques for domain-specific applications.					L4, L5, L6	
Semester	Autumn: Yes			Spring: No			
	Lecture	Tutorial	Practical		Credits	Total teaching hours	
Contact Hours	3	0	2		4	60	
Prerequisite course code as per proposed course numbers	Machine Learning Course						
Prerequisite credits	NIL						
Equivalent course codes as	NIL						

per proposed course and old course					
Overlap course codes as per proposed course numbers	NIL				
TextBooks					
1	Title	Deep Learning: Foundations and Concepts			
	Author	Christopher M. Bishop & Hugh Bishop			
	Publisher	Springer			
	Edition	2023			
2	Title	Deep Learning			
	Author	Ian Goodfellow and Yoshua Bengio and Aaron Courville			
	Publisher	MIT Press			
	Edition	2016			
Reference Books					
1	Title	Machine Learning: An Algorithmic Perspective, Second Edition			
	Author	Stephen Marsland			
	Publisher	Chapman and Hall/CRC			
	Edition	2nd			
2	Title	Introduction to Probability For Data Science			
	Author	Stanley H. Chan			
	Publisher	Michigan Publishing			
	Edition	May 2021			
Content	Unit 1: Introduction to Deep Learning (10 Hours) Overview of Deep Learning and its comparison with Machine Learning, Artificial Neural Networks (ANNs): Structure, perceptrons, limitations, forward propagation, backpropagation, activation functions (e.g., ReLU, Softmax), loss functions, optimization techniques (e.g., SGD, Adam), Regularization techniques: L1/L2 regularization, Dropout, Batch Normalization, Hyperparameter tuning, Commonly used frameworks (e.g., TensorFlow, PyTorch). Unit 2: Convolutional Neural Networks (CNNs) (9 Hours) Key concepts: convolution, padding, pooling, CNN layers, Architectures: AlexNet, VGG, ResNet, MobileNet, and others, Applications: Image classification, transfer learning, object detection (e.g., YOLO, R-CNN),				

	encoder-decoder models, Image segmentation: semantic, instance, and panoptic segmentation using U-Net and its variants, Regularization methods, Data augmentation techniques. Unit 3: Recurrent Neural Networks (RNNs) (6 Hours) RNN architecture and working principles, Variants: LSTM, GRU, Advantages in handling sequential data, Challenges: vanishing and exploding gradients, Applications: NLP tasks such as sentiment analysis, text generation, sequence prediction. Unit 4: Transformers (6 Hours) Transformer architecture: encoder-decoder model, self-attention, multi-head attention, Applications: NLP tasks such as translation, summarization, sentiment analysis, Vision Transformers (ViTs), Swin Transformer and its hierarchical representation learning, Large Language Models (LLMs): BERT, GPT and their applications. Unit 5: Advanced Topics (5 Hours) Autoencoders and Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs): architecture and applications, Diffusion models in generative AI, Time Series Forecasting: ARIMA, SARIMA, deep learning approaches (LSTM, GRU), Federated Learning: privacy-preserving distributed training, Real-world implementations in speech and gesture recognition.
Course Assessment	Continuous Evaluation 25%
	Mid Semester 25%
	End Semester 50%

Course Matrix (CO-PO-PSO Mapping)

COs	POs s & PSOs													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	2			2							1	3	2
CO2	3	3	3	2	3							1	3	3
CO3	3	3	3	2	3							1	3	3
CO4	3	3	3	3	3							2	3	3

1=addressed to small extent

2= addressed significantly

3= addressed strongly (major part of course)

Lab Experiments

Exp No.	List of Experiments
1	MNIST Digit Classification: Train a simple feedforward neural network on the MNIST dataset using forward and backpropagation.
2	Image Classification with CNN: Develop and evaluate a CNN model on datasets like CIFAR-10 or Fashion-MNIST.
3	Transfer Learning with Pre-trained CNN: Fine-tune models like VGG16, ResNet, or MobileNet on a custom image dataset.
4	Object Detection using YOLO: Implement YOLOv5 or YOLOv8 for real-time object detection on images or video frames.
5	Semantic Segmentation with U-Net: Use the U-Net architecture for medical or satellite image segmentation.
6	Instance Segmentation with Mask R-CNN: Apply Mask R-CNN for pixel-wise object segmentation on COCO or similar datasets.
7	Image Classification using ViT and Swin Transformer: Train Vision Transformer (ViT) and Swin Transformer models, and compare their performance with CNNs on image classification tasks.
8	Diffusion Model for Image Generation: Use a pretrained or simplified diffusion model (e.g., Stable Diffusion) to generate high-quality synthetic images.
9	Autoencoder and Variational Autoencoder (VAE): Build autoencoders for image denoising and extend to VAEs for generative modeling.
10	Generative Adversarial Networks (GANs): Implement a basic GAN or DCGAN to generate synthetic image data.
11	Time Series Forecasting with LSTM and SARIMA: Compare LSTM-based forecasting with classical SARIMA for time series prediction tasks.
12	Sentiment Analysis using LSTM and BERT: Perform sentiment classification using LSTM and fine-tuned BERT on datasets like IMDB or Twitter.

Course Code: ADBB 302	PC (YES/ NO)	PE (YES/ NO)	OE (YES/ NO)	AS (YES/ NO)	HM (YES / NO)	ST-IS-PR (YES/ NO)	AE (YES / NO)
	YES	NO	NO	NO	NO	NO	NO
Type of course	Program Core						
Course Title	NATURAL LANGUAGE PROCESSING						
Course Objectives	To provide a broad introduction to NLP with a particular emphasis on core algorithms, data structures, and machine learning for NLP, text classification, sentiment analysis and other applications of NLP.						
Course Outcomes	CO1: Understand the fundamental concepts of natural language processing, including models, ambiguity, processing paradigms, and phases of NLP along with text representation in computers.					L1, L2, L3	
	CO2: Apply linguistic resources and tools such as corpus, WordNet, TreeBank, and Finite State Automata to analyze morphology and word recognition using probabilistic models like N-grams and HMM.					L4, L5, L6	
	CO3: Demonstrate the ability to perform Part-of-Speech tagging, statistical and probabilistic parsing, and handle challenges like unknown words and multi-word expressions.					L4, L5, L6	
	CO4: Design semantic analysis techniques, Word Sense Disambiguation methods, and NLP applications such as sentiment analysis, summarization, and machine translation.					L4, L5, L6	
Semester	Autumn: Yes			Spring: No			
	Lecture	Tutorial	Practical		Credits	Total teaching hours	
Contact Hours	3	1	0		4	48	
Prerequisite course code as	Machine Learning						

per proposed course numbers	Course				
Prerequisite credits	NIL				
Equivalent course codes as per proposed course and old course	NIL				
Overlap course codes as per proposed course numbers	NIL				
TextBooks					
1	Title	Speech and Language Processing			
	Author	Daniel Jurafsky and James H Martin			
	Publisher	Pearson Education			
	Edition	2009			
2	Title	Natural language processing and Information retrieval			
	Author	Siddiqui T., Tiwary U. S.			
	Publisher	OUP			
	Edition	2008			
Reference Books					
1	Title	Natural language Understanding			
	Author	James A			
	Publisher	Pearson Education			
	Edition	1994			
2	Title	Natural language processing: a Paninian perspective			
	Author	Bharati A., Sangal R., Chaitanya V.			
	Publisher	PHI			
	Edition	2000			

Content	Unit 1: Introduction to Natural Language Processing (8 Hours) Human and Natural Language, Features of Language, Ambiguity in Language, NLP Applications, Rule-Based NLP, Statistical NLP, Machine Learning Approaches, Morphological Analysis, Syntactic Analysis, Semantic Analysis, Text Representation in Computers, ASCII, Unicode, Bag of Words, TF-IDF, Word Embeddings, NLP Tools and Libraries (NLTK, SpaCy). Unit 2: Linguistic Resources and Statistical Foundations (9 Hours) Corpus, Balanced Corpus, TreeBank, PropBank, FrameNet, WordNet, VerbNet, Regular Expressions, Finite State Automata, Tokenization, Lexical Analysis, Morphology, Inflection, Derivation, Finite State Transducer, n-gram Language Models, Smoothing Techniques, Entropy, Perplexity, Hidden Markov Models (HMM), Maximum Entropy Models. Unit 3: POS Tagging and Parsing Techniques (9 Hours) Part-of-Speech Tagging, Rule-Based Tagging, Stochastic Tagging, HMM Tagging, Transformation-Based Learning (TBL), Unknown Words, Named Entities, Multiword Expressions, Syntax Trees, Dependency Parsing, CYK Algorithm, Earley Parser, CKY Parser, Transition-Based Parsing, Statistical Parsing, Probabilistic Parsing. Unit 4: Semantics and Text Analysis (5 Hours) Lexical Semantics, Synonymy, Polysemy, WordNet, Word Sense Disambiguation, Dictionary-Based WSD, Supervised WSD, Unsupervised WSD, Predicate Logic, Semantic Role Labelling, Text Classification, Text Clustering, Sentiment Analysis, Subjectivity Detection. Unit 5: NLP Applications (5 Hours) Spell Checking, Text Normalization, Text Summarization, Extractive Summarization, Abstractive Summarization, Machine Translation, Rule-Based MT, Statistical MT, Neural MT, Chatbots, Dialogue Systems, Question Answering, Information Retrieval, Text Generation, Named Entity Recognition, Speech Recognition, Text-to-Speech (TTS), Topic Modelling, Latent Dirichlet Allocation (LDA).
Course Assessment	Continuous Evaluation 25% Mid Semester 25% End Semester 50%

Course Matrix (CO-PO-PSO Mapping)

COs	POs s & PSOs													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2

CO1	3	2			2							1	3	2
CO2	3	3	2	2	3							1	3	2
CO3	3	3	2	2	3							1	3	3
CO4	3	3	2	3	3							2	3	3

1=addressed to small extent

2= addressed significantly

3= addressed strongly (major part of course)

Course Code: ADBB 303	PC (YES/ NO)	PE (YES/ NO)	OE (YES/ NO)	AS (YES/ NO)	HM (YES / NO)	ST-IS-PR (YES/ NO)	AE (YES / NO)
	YES	NO	NO	NO	NO	NO	NO
Type of course	Program Core						
Course Title	CLOUD COMPUTING						
Course Objectives	The course aims to introduce the fundamentals of cloud computing, including its architecture, service models (IaaS, PaaS, SaaS), and deployment models. It equips students with knowledge of virtualization and cloud storage along with hands-on experience using leading cloud platforms.						
Course Outcomes	CO1: Understand the fundamental concepts, service models, and deployment strategies of cloud computing and evaluate its benefits and challenges in real-world applications.					L2	
	CO2: Analyze the architecture of cloud systems and apply knowledge of virtualization and data center technologies to design scalable and efficient cloud solutions.					L4	
	CO3: Implement and evaluate basic cloud-based programs using distributed programming frameworks.					L5	
	CO4: Develop and demonstrate the ability to use modern cloud platforms and services for application development and deployment.					L6	
Semester	Autumn: Yes			Spring:			
	Lecture	Tutorial	Practical		Credits	Total teaching hours	
Contact Hours	3	0	2		4	60	
Prerequisite course code as per proposed course numbers	NIL						
Prerequisite credits	NIL						

[illegible]

CO3	2	2	3		2								3	2
CO4	2	2	3	1	3								3	1

1=addressed to small extent

2= addressed significantly

3= addressed strongly (major part of course)

Lab Experiments

Exp No.	List of Experiments
1	Install hypervisor with linux or windows OS on top of host OS.
2	Install a C compiler in the virtual machine created using virtual box and execute Simple Programs.
3	Install Google App Engine (GAE). Create hello world app and other simple web applications using python/java.
4	Use Google App Engine (GAE) launcher to launch the web applications.
5	Simulate a cloud scenario using CloudSim and run a scheduling algorithm that is not present in CloudSim.
6	Implement a procedure to transfer the files from one virtual machine to another virtual machine.
7	Install Hadoop node cluster and run basic applications.

Course Code: ADLB 304	PC (YES/ NO)	PE (YES/ NO)	OE (YES/ NO)	AS (YES/ NO)	HM (YES / NO)	ST-IS-PR (YES/ NO)	AE (YES / NO)
	YES	NO	NO	NO	NO	NO	NO
Type of course	Program core						
Course Title	IMAGE PROCESSING AND COMPUTER VISION						
Course Objectives	To provide students with a comprehensive understanding of digital image processing techniques, including spatial and frequency domain transformations, image segmentation, and color image processing. The course aims to enable students to develop image processing tools from scratch and gain hands-on experience using OpenCV and MATLAB.						
Course Outcomes	CO1: Learn the basics and mathematical background of Image Processing					L1, L3	
	CO2: Analysis and study of methods used for image sampling and quantization, image transforms, image enhancement and restoration, image encoding, image analysis and pattern recognition.					L2, L4	
	CO3: Utility of image compression techniques for storage and transmission purpose					L3, L5	
	CO4: To learn about color imaging, color models, and color image processing.					L4	
Semester	Autumn: Yes			Spring:			
	Lecture	Tutorial	Practical		Credits	Total teaching hours	
Contact Hours	3	0	2		4	60	
Prerequisite course code as per proposed course numbers	NIL						
Prerequisite	NIL						

credits					
Equivalent course codes as per proposed course and old course	NIL				
Overlap course codes as per proposed course numbers	NIL				
TextBooks					
1	Title	Digital Image Processing			
	Author	R.C. Gonzalez, R.E Woods			
	Publisher	Pearson Education			
	Edition	2008			
Reference Books					
1	Title	Digital Image Processing Using MATLAB			
	Author	R.C. Gonzalez, R.E Woods, S. L. Eddins			
	Publisher	PHI			
	Edition	2003			
2	Title	Image Processing, Analysis, and Machine Vision			
	Author	M. Sonka, V. Hlavac, R. Boyle			
	Publisher	Brooks/Cole			
	Edition	2007			
3	Title	Digital Image Processing			
	Author	W.K. Pratt			
	Publisher	Wiley-Interscience			
	Edition	2007			
Content	Unit 1: Digital Image Fundamentals (9 Hours) Introduction: Digital image representation, Fundamental steps in image processing, Components of Digital Image processing systems, Elements of visual perception, Image Formation model, Image Sampling and quantization, Relationship between pixels-neighbourhood, adjacency connectivity, regions, boundaries and distance measures. Unit 2: Image Enhancement Techniques (9 Hours) Image Enhancement: Enhancement by point processing, Sample intensity				

	transformation, Histogram processing, Image subtraction, Image averaging, Spatial Filtering-Smoothing Spatial filters, Sharpening Spatial filters, Frequency domain-Fourier Transform, Low-Pass, High-Pass, Laplacian, Homomorphic filtering. Unit 3: Image Segmentation Methods (9 Hours) Image Segmentation: Detection of discontinuities – point, line and edge detection, Edge linking and boundary detection, Thresholding, Region-based segmentation – region growing, region splitting and merging, Use of motion in segmentation- Spatial techniques and Frequency domain techniques. Unit 4: Image Compression Principles (5 Hours) Image Compression: Coding redundancy, Interpixel redundancy, fidelity criteria, Image compression models, Error-free compression, Variable length coding, Bit-plane coding, Lossless predictive coding, Lossy compression, Image compression standards, Real-Time image transmission, JPEG and MPEG. Unit 5: Color Image Processing (4 Hours) Color Image Processing: Color Models, Pseudo color Image Processing, Color Transformations, Smoothing and sharpening, Image Segmentation based on color.
Course Assessment	Continuous Evaluation 25%
	Mid Semester 25%
	End Semester 50%

Course Matrix (CO-PO-PSO Mapping)

COs	POs s & PSOs													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	2			2							1	3	2
CO2	3	3	3	2	3							1	3	3
CO3	3	2	3	2	3							2	3	3
CO4	2	2	2	2	2							1	2	2

1=addressed to small extent

2= addressed significantly

3= addressed strongly (major part of course)

Lab Experiments

Exp No.	List of Experiments
1	Image Acquisition and Display: Read and display grayscale and color images using MATLAB or Python.
2	Gray-Level Transformations: Apply negative, log, and gamma correction transformations.
3	Histogram Equalization: Enhance contrast using histogram equalization.
4	Noise Removal Using Filters: Apply mean and median filters to remove salt-and-pepper noise.
5	Edge Detection: Detects edges using Sobel, Canny, and Laplacian operators.
6	Fourier Transform and Filtering: Perform FFT, apply low-pass and high-pass filtering, and reconstruct the image.
7	Image Restoration: Simulate degradation and restore using Wiener filtering.
8	Color Image Processing: Convert RGB to HSI and display individual components.
9	Morphological Operations: Perform dilation, erosion, opening, and closing on binary images.
10	Image Segmentation: Apply thresholding, edge-based, and watershed segmentation.

Course Code: ADBB 305	PC (YES/ NO)	PE (YES/ NO)	OE (YES/ NO)	AS (YES/ NO)	HM (YES / NO)	ST-IS-PR (YES/ NO)	AE (YES / NO)
	YES	NO	NO	NO	NO	NO	NO
Type of course	Program core						
Course Title	INTERNET OF THINGS						
Course Objectives	To provide a broad introduction to the Internet of Things (IoT), focusing on its enabling technologies, hardware components, and practical applications. The course aims to help students understand the evolution, definition, and scope of IoT, and its role in modern technology. It also covers the study of sensors, actuators, microcontrollers (such as Arduino and Raspberry Pi), and embedded systems used in IoT development.						
Course Outcomes	CO1: Understand the fundamentals of the Internet of Things, including its architecture, characteristics, enabling technologies, and real-world applications such as smart homes, healthcare, and smart cities.					L1, L2	
	CO2: Identify and utilize appropriate IoT hardware components, microcontrollers, and communication protocols to design basic IoT systems.					L3	
	CO3: Analyze the integration of IoT systems with cloud platforms and apply concepts of cloud storage, edge, and fog computing for IoT data processing and analytics.					L4	
	CO4: Evaluate and implement basic security and privacy measures in IoT systems and develop small-scale IoT applications using Python and open-source platforms like Arduino or Raspberry Pi.					L5, l6	
Semester	Autumn: Yes			Spring:			
	Lecture	Tutorial	Practical		Credits	Total teaching hours	
Contact Hours	3	0	2		4	60	
Prerequisite course code as	NIL						

per proposed course numbers					
Prerequisite credits	NIL				
Equivalent course codes as per proposed course and old course	NIL				
Overlap course codes as per proposed course numbers	NIL				
TextBooks					
1	Title	Introducing Internet of Things			
	Author	Arshdeep Bahga, Vijay Madisetti			
	Publisher	Pearson			
	Edition	1st Edition, 2015			
Reference Books					
1	Title	Introducing Internet of Things			
	Author	Raj Kamal			
	Publisher	McGraw Hill Education			
	Edition	1st Edition, 2017			
2	Title	Designing the Internet of Things			
	Author	Adrian McEwen, Hakim Cassimally			
	Publisher	Wiley			
	Edition	1st Edition, 2014			
3	Title	Internet of Things: a Modern Approach			
	Author	Olivier Hersent, David Boswarthick, Omar Elloumi			
	Publisher	Wiley			
	Edition	2nd Edition, 2016			

Content	Unit 1: Introduction to IoT (6 Hours) Definition, Characteristics, and Evolution of IoT, IoT Ecosystem & Architecture, Physical design and Logical design of IoT, Enabling Technologies: RFID, Bluetooth, Zigbee, GPS, Wi-Fi, 4G/5G, Challenges and Applications of IoT (Smart Home, Smart City, Healthcare, etc). Unit 2: IoT Hardware and Communication (8 Hours) IoT Devices: Sensors, Actuators, Microcontrollers (e.g., Arduino, Raspberry Pi), Embedded systems overview, Communication Models and APIs, IoT Communication Protocols: MQTT, CoAP, AMQP, HTTP, XMPP, M2M Communication. Unit 3: IoT Platforms and Cloud Integration (7 Hours) IoT and Cloud Computing, Introduction to IoT Platforms: Google Cloud IoT, AWS IoT, Azure IoT, Cloud Storage Models for IoT Data, Data Analytics in IoT, Edge and Fog Computing in IoT. Unit 4: Security and Privacy in IoT (7 Hours) IoT Security Challenges, Authentication and Authorization in IoT, IoT Security Architecture, Common Attacks in IoT and their Mitigation, Privacy Preservation in IoT. Unit 5: IoT Application Development (8 Hours) IoT System Design Methodology, Developing IoT Applications using Python, Hands-on with Arduino/Raspberry Pi, Case Studies: Smart Agriculture, Smart Healthcare, Smart Energy, etc, Mini Project / Lab Implementation.
Course Assessment	Continuous Evaluation 25% Mid Semester 25% End Semester 50%

Course Matrix (CO-PO-PSO Mapping)

COs	POs s & PSOs													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	2			2							1	2	2
CO2	3	2	3		3							1	3	3
CO3	3	3	3	3	3							2	3	3
CO4	3	3	3	3	3							2	3	3

1=addressed to small extent

2= addressed significantly

3= addressed strongly (major part of course)

Lab Experiments

Exp No.	List of Experiments
1	Introduction to Arduino and Raspberry Pi: LED Blinking using GPIO
2	Sensor Data Acquisition: Reading Temperature and Humidity Using Sensor
3	Actuator Control: Automating Fan/LED based on Sensor Input
4	IoT Communication: Publishing Sensor Data using MQTT Protocol
5	Remote Monitoring: Sending Sensor Data using HTTP Requests
6	Cloud Integration: Real-time Data Logging on IoT Platforms (ThingSpeak/AWS IoT)
7	IoT Security: User Authentication and Data Encryption in IoT Applications
8	Smart Home Simulation: Controlling Devices via Smartphone App
9	Smart Agriculture System: Automated Irrigation using Soil Moisture Sensor

Course Code: ADLB 351	PC (YES/ NO)	PE (YES/ NO)	OE (YES/ NO)	AS (YES/ NO)	HM (YES / NO)	ST-IS-PR (YES/ NO)	AE (YES / NO)
	YES	NO	NO	NO	NO	NO	NO
Type of course	Program core						
Course Title	SOCIAL NETWORK ANALYTICS						
Course Objectives	To introduce the fundamentals and graph-based representation of social networks, study key models and measures for analysis, explore techniques for community detection and link prediction, understand diffusion and influence models in social media, and develop skills for mining and visualizing social media data, with attention to ethical considerations.						
Course Outcomes	CO1: Understand the basic structure of social networks, their types, graph-based representations, and foundational graph theory concepts.					L1, L2	
	CO2: Analyze and apply network models and centrality/structural measures to characterize and evaluate social networks.					L3, L4	
	CO3: Apply and evaluate community detection and link prediction algorithms in social networks, using appropriate evaluation metrics.					L4, L5	
	CO4: Design and implement solutions for analyzing influence propagation, mining and visualizing social media data, while addressing privacy and ethical concerns.					L5, l6	
Semester	Autumn:			Spring: Yes			
	Lecture	Tutorial	Practical		Credits	Total teaching hours	
Contact Hours	3	1	0		4	48	
Prerequisite course code as per proposed course numbers	NIL						
Prerequisite credits	NIL						

Equivalent course codes as per proposed course and old course	NIL				
Overlap course codes as per proposed course numbers	NIL				
TextBooks					
1	Title	Introducing Social Network Analytics			
	Author	Matthew A. Russell			
	Publisher	O'Reilly Media			
	Edition	3rd Edition, 2018			
Reference Books					
1	Title	Introducing Social Network Analysis			
	Author	Stanley Wasserman & Katherine Faust			
	Publisher	Cambridge University Press			
	Edition	1st Edition, 1994			
2	Title	Networks, Crowds, and Markets			
	Author	David Easley & Jon Kleinberg			
	Publisher	Cambridge University Press			
	Edition	1st Edition, 2010			
3	Title	Social Network Analysis for Startups			
	Author	Maksim Tsvetovat & Alexander Kouznetsov			
	Publisher	O'Reilly Media			
	Edition	1st Edition, 2011			

Content	Unit 1: Introduction to Social Networks Analytics (6 Hours) Definition and types of social networks (online, offline, ego networks), Real-life applications: Facebook, Twitter, LinkedIn, citation networks, Structure of social networks: Nodes, edges, adjacency matrices, Basic concepts in graph theory: Paths, connectivity, degree, clustering coefficient. Unit 2: Social Network Analytics Models and Measures (8 Hours) Random network models: Erdős–Rényi model, Small-world networks: Watts-Strogatz model, Scale-free networks: Barabási–Albert model, Centrality measures: Degree, Closeness, Betweenness, Eigenvector, Network density, Diameter, Reciprocity, Assortativity. Unit 3: Social Network Community Detection and Link Prediction (8 Hours) Communities and modularity, Algorithms: Girvan-Newman, Louvain Method, Label Propagation, Link prediction methods: Similarity-based, probabilistic models, Evaluation metrics: Precision, recall, AUC. Unit 4: Social Network Information Diffusion and Influence Analysis (7 Hours) Diffusion models: Independent Cascade Model, Linear Threshold Model, Influence maximization: Greedy algorithms, Contagion models: Viral marketing, epidemic modeling, Case study: Twitter hashtag propagation. Unit 5: Mining and Visualization of Social Media Data (7 Hours) Text mining from social networks, Sentiment analysis and opinion mining, Social media APIs (Twitter, Facebook Graph API), Visualization tools: Gephi, NetworkX, Cytoscape, Ethics, privacy, and challenges in social media analytics.
Course Assessment	Continuous Evaluation 25%
	Mid Semester 25%
	End Semester 50%

Course Matrix (CO-PO-PSO Mapping)

COs	POs s & PSOs													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	2			2							1	2	2
CO2	3	3	3	2	3							1	3	3
CO3	3	3	3	3	3							2	3	3
CO4	3	3	3	3	3							2	3	3

1=addressed to small extent

2= addressed significantly

3= addressed strongly (major part of course)

Course Code: ADBB 352	PC (YES/ NO)	PE (YES/ NO)	OE (YES/ NO)	AS (YES/ NO)	HM (YES / NO)	ST-IS-PR (YES/ NO)	AE (YES / NO)
	YES	NO	NO	NO	NO	NO	NO
Type of course	Program Core						
Course Title	BIG DATA ANALYTICS						
Course Objectives	This course aims to provide the fundamental concepts of Big Data and Analytics; explore tools and practices for working with Big Data and stream computing; understand the Big Data use cases; apply analytics on structured and unstructured data with R; and provide comprehensive knowledge on developing and applying Machine Learning algorithms for massive real-world datasets in distributed frameworks.						
Course Outcomes	CO1: Explain the concept and significance of Big Data and its analysis.					L2	
	CO2: Apply and Analyze regression and classification algorithms for Big Data analytics.					L4	
	CO3: Analyze different mining algorithms and clustering techniques for Big Data Analytics.					L4	
	CO4: Design and develop big data-based analytics for real-word ubiquitous computing scenarios.					L6	
Semester	Autumn:			Spring: Yes			
	Lecture	Tutorial	Practical		Credits	Total teaching hours	
Contact Hours	3	0	2		4	60	
Prerequisite course code as per proposed course numbers	NIL						
Prerequisite credits	NIL						
Equivalent course	NIL						

codes as per proposed course and old course					
Overlap course codes as per proposed course numbers	NIL				
TextBooks					
1	Title	Big Data Analytics: From Strategic Planning to Enterprise Integration with Tools, Techniques, NoSQL, and Graph			
	Author	David Loshin			
	Publisher	Morgan Kaufmann/Elsevier Publishers			
	Edition	2013			
2	Title	Mining of Massive Datasets			
	Author	Anand Rajaraman and Jeffrey David Ullman			
	Publisher	Cambridge University Press			
	Edition	2012			
Reference Books					
1	Title	Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data			
	Author	EMC Education Services			
	Publisher	Wiley			
	Edition	2015			
2	Title	Analytics in a Big Data World: The Essential Guide to Data Science and its Applications			
	Author	Beasan Bart			
	Publisher	Wiley			
	Edition	2015			
Content	Unit 1: Introduction to Big Data (7 Hours) Evolution of big data, best practices for big data analytics, big data characteristics; Validating, Promotion of the value of big data, big data use cases. Characteristics of big data applications, Perception and quantification of value. Big Data Tools and Techniques: Understanding big data storage, General overview of high-performance architecture, HDFS, Map Reduce and YARN; Map Reduce programming model; Review of basic data analytic methods using R.				

	Unit 2: Regression and Classification (7 Hours) Advanced analytical theory and methods, Regression: Linear regression, Logistic regression; Classification: Decision trees, Overview of a decision tree, Decision tree algorithms, Evaluating a decision tree, Decision trees in R, Naïve Bayes, Bayes 'theorem, Naïve Bayes classifier in R. Unit 3: Data Stream Analysis (7 Hours) Introduction to streams concepts: Stream data model and architecture, Stream computing, Sampling data in a stream, filtering streams, counting distinct elements in a stream, estimating moments, counting oneness in a window, Decaying window; Real Time Analytics Platform (RTAP) applications, Case studies: Real time sentiment analysis, Stock market predictions. Unit 4: Frequent Itemset and Clustering (7 Hours) Mining frequent itemset: Market based model, Apriori algorithm, handling large datasets in main memory, Limited Pass algorithm, counting frequent itemset in a stream, Clustering techniques: Hierarchical, k-Means, Clustering high dimensional data. Unit 5: NoSQL Data Management for Big Data (8 Hours) NoSQL databases: Schema-less models, increasing flexibility for data manipulation, Key value stores, Document stores, Tabular stores, Object data stores, Graph databases; Hive; Sharding; HBase; Case Study: Analyzing big data with twitter, big data for E-Commerce Big data for blogs.
Course Assessment	Continuous Evaluation 25%
	Mid Semester 25%
	End Semester 50%

Course Matrix (CO-PO-PSO Mapping)

COs	POs s & PSOs													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	2	2	3	1	1			3					1	2
CO2	2	3	3	3	1							3	3	2
CO3	2	3	3	3	1							3	3	2
CO4	2	2	3	3	1			3				3	3	3

1=addressed to small extent

2= addressed significantly

3= addressed strongly (major part of course)

Lab Experiments

Exp No.	List of Experiments
1	Study and Configuration Hadoop-based distributed architecture for Big Data Analytics.
2	Map Reduce Programming Examples Word Count. Union, Intersection and Difference. Matrix Multiplication.
3	Installation and Creation of MongoDB for Schema-less database.
4	Study and Implementation of Regression-based algorithms for Big Data Analytics.
5	Study and Implementation of Clustering-based algorithms for Big Data Analytics.
6	Implement and Perform Streaming Data Analysis for X data (formerly twitter), chat data, weblog analysis.
7	Implementation of Visualization techniques for Interpreting Big Data and its Analytics.

Course Code: ADBB 353	PC (YES/ NO)	PE (YES/ NO)	OE (YES/ NO)	AS (YES/ NO)	HM (YES / NO)	ST-IS-PR (YES/ NO)	AE (YES / NO)
	YES	NO	NO	NO	NO	NO	NO
Type of course	Program Core						
Course Title	SOFT COMPUTING						
Course Objectives	To provide an understanding of the fundamental concepts of neural networks, fuzzy logic, and genetic algorithms, along with their applications in soft computing.						
Course Outcomes	CO1: Understand the fundamentals of artificial neural networks, including early models like ADALINE and MADALINE, and different neural network architectures such as feedforward and recurrent networks.					L2	
	CO2: Apply supervised and unsupervised learning techniques to build and analyze different neural network models, including associative memory networks and Hopfield networks.					L4	
	CO3: Analyze fuzzy logic concepts such as fuzzy sets, fuzzy arithmetic, fuzzy rule-based systems, and defuzzification techniques to develop fuzzy inference systems.					L4	
	CO4: Evaluate the role of genetic algorithms and hybrid intelligent systems (e.g., neuro-fuzzy, genetic-fuzzy, and genetic-neural hybrids) in solving optimization and learning problems.					L6	
Semester	Autumn:			Spring: Yes			
	Lecture	Tutorial	Practical		Credits	Total teaching hours	
Contact Hours	3	0	2		4	60	
Prerequisite course code as per proposed course numbers	NIL						
Prerequisite	NIL						

credits					
Equivalent course codes as per proposed course and old course	NIL				
Overlap course codes as per proposed course numbers	NIL				
TextBooks					
1	Title	A comprehensive foundation. Neural Networks			
	Author	Simon Haykin			
	Publisher	Pearson Education			
	Edition	2nd Edition, 2001.			
Reference Books					
1	Title	Fuzzy logic with engineering applications			
	Author	Timothy J. Ross			
	Publisher	John Wiley & Sons			
	Edition	3rd Edition,2009			
2	Title	An Introduction to Genetic Algorithms			
	Author	Melanie Mitchell			
	Publisher	Prentice-Hall			
	Edition	1998			
3	Title	Genetic Algorithms in Search, Optimization, and Machine Learning			
	Author	D. E. Goldberg			
	Publisher	Addison-Wesley			
	Edition	1989			
Content	Unit 1: Introduction to Artificial Neural Networks (7 Hours) Artificial Neural Networks: Basic concepts of artificial neural networks, earlier neural networks: ADALINE, MADALINE. Neural Network Architectures: Single layer feedforward network, Multi-layer feedforward network, Recurrent network. Unit 2: Learning Networks and Associative Memory (11 Hours) Supervised Learning Network: Perceptron network, Back propagation network,				

	Radial basis function network. Unsupervised Learning Network: Fixed weight competitive nets, Kohonen self-organizing feature maps, Counter propagation network, Adaptive reasoning theory. Associative memory: Auto-associative memory network, Hetero Associative memory network, Bidirectional associative memory, Hopfield networks. Unit 3: Fuzzy Logic and Fuzzy Systems (10 Hours) Fuzzy Logic: Crisp set and Fuzzy set, Basic concepts of fuzzy sets, Fuzzy set operations, Fuzzy Arithmetic-fuzzy numbers, Fuzzy ordering, Fuzzy vectors. Fuzzy measures-belief and plausibility measure. Probability measure: Measure of fuzziness, Fuzzy integrals. Membership functions: Features of membership function, Fuzzification. Fuzzy Rule Based Systems: Fuzzy proposition, Formation and decomposition of rules, Fuzzy reasoning, Fuzzy inference systems, Fuzzy expert system. Defuzzification: Max-membership, Centroid method, Weighted average, Mean max. Unit 4: Genetic Algorithms and Search Techniques (8 Hours) Genetic Algorithms: Traditional optimization and search techniques, Genetic algorithms. Operators: Encoding, Selection, Crossover, Mutation. Classification: Adaptive genetic algorithms, Hybrid genetic algorithms, Parallel genetic algorithms, Real coded genetic algorithms.
Course Assessment	Continuous Evaluation 25%
	Mid Semester 25%
	End Semester 50%

Course Matrix (CO-PO-PSO Mapping)

COs	POs s & PSOs													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	2			2							1	3	2
CO2	3	3	3	2	3							1	3	3
CO3	3	3	3	2	3							1	3	3
CO4	3	3	3	3	3							2	3	3

1=addressed to small extent

2= addressed significantly

3= addressed strongly (major part of course)

Lab Experiments

Exp No.	List of Experiments
1	Train a single-layer perceptron to classify linearly separable data.
2	Demonstrate learning and convergence for simple classification tasks.
3	Train a neural network for XOR or digit classification problems using backpropagation.
4	Design and test an RBF network on a small dataset.
5	Implement SOM for clustering or dimensionality reduction.
6	Use a Hopfield network to store and recall binary patterns.
7	Perform fuzzy union, intersection, and complement operations.
8	Create a Mamdani-type or Sugeno-type FIS for a real-world problem (e.g., temperature control).
9	Compare different defuzzification methods (centroid, mean of maxima, etc.) for a fuzzy output.
10	Solve a mathematical optimization problem using a basic GA with crossover and mutation.